

Curriculum Vitae Dr. Fabrice Lacroix

Citizenships	Switzerland, Canada, France
Date Of Birth	07.09.1989
Languages	English (native-speaker), German (native-speaker), French (native-speaker)
ORCID	https://orcid.org/0000-0003-4749-2826
Google Scholar ID	https://scholar.google.de/citations?user=8mIU_wQAAAAJ&hl=de
Personal Website	https://fabricelacroix.github.io
Ambizione Webpage:	https://www.geography.unibe.ch/research/geocomputation_and_earth_observation/high_latitude_biogeochemical_feedbacks/index_eng.html

Education and Degrees

07.2019	Ph.D. in Science <i>Université Libre de Bruxelles, Brussels, Belgium</i> <i>Max-Planck-Institute for Meteorology, Hamburg, Germany</i> Dissertation title: " Global and regional contributions of riverine fluxes to the global and regional oceanic cycling of carbon and nutrients " Advisors: Prof. Dr. Pierre Regnier, Prof Dr. Tatiana Ilyina
07.2015	M.Sc. in Environmental Science <i>ETH Zürich, Zürich, Switzerland</i> Thesis title: <i>Offshore transport of carbon in the California current system</i>
10.2013	B.Sc. in Environmental Science <i>ETH Zürich, Zürich, Switzerland</i>

Current Position

03.2024 –	Ambizione Fellow - Senior Research Assistant <i>Institute of Geography & Oeschger Centre for Climate Change Research, University of Bern, Switzerland</i> - Project lead (1 PhD student), managing finances, research output and human resources
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- Analysis of past and future permafrost thaw for Arctic greenhouse gas exchange
- Co-development of the terrestrial biosphere model QUINCY-(ICON-Land) and coastal model ICON-COAST
- Participation in international greenhouse assessments (RECCAP2 assessments)

Past Academic Positions

- 04.2022 – 01.2024 Postdoctoral Researcher, H2020 Project PROVIDE**
Climate and Environmental Physics & Oeschger Centre for Climate Change Research, University of Bern, Switzerland
- Co-development of the Adaptive Emission Reduction Approach to simulate overshoot scenarios in the coupled Earth System model GFDL-ESM2M
 - Investigation of Earth System reversibility in overshoot scenarios
 - Co-development of PROVIDE Climate Risk Dashboard
- 09.2023 – 11.2023 Guest Researcher (2.5 months), UniBe Travel Grant Recipient**
Laboratory of Climate Extremes, Universidade Federal de Santa Catarina, Brazil
- Analysis of overshoot impacts in the South Atlantic region
- 04.2020 – 04.2022 Research Associate, Postdoctoral Researcher**
Department Biogeochemical Signals, Max-Planck-Institute for Biogeochemistry, Jena, Germany
- Development of snow and freeze-thaw modules in the model QUINCY
 - Investigation of past trends in high-Arctic vegetation growth and greenhouse gas emissions using the QUINCY model and observations
- 07.2019 – 03.2020 Postdoctoral Researcher**
Department Ocean in the Earth System, Max-Planck-Institute for Meteorology, Hamburg, Germany
- Development of snow and freeze-thaw modules in the model ICON-LAND-QUINCY
 - Conducted investigation of past trends in high-Arctic vegetation growth and greenhouse gas emissions using the QUINCY model and observations
- 09.2015 – 07.2019 EU H2020 Marie Skłodowska-Curie Early-Stage Researcher**
Université Libre de Bruxelles, Brussels, Belgium / Max-Planck-Institute for Meteorology, Hamburg, Germany
- Development of river and coastal ocean model representations in Earth system model MPI-ESM
 - Outreach to multidisciplinary stakeholders

Other Employment

03.2015 – 06.2015	Intern Climate Accounting <i>Climate-KIC, Zürich, Switzerland</i>
11.2014 – 02.2015	Intern Casualty Risk Engineering <i>XL Insurance (Acquired by Catlin Insurance), Zürich Switzerland</i>

Teaching

09.2024 –	Co-Lecturer for " Land in the Earth System II " - M.Sc. Climate Science Graduate course, <i>University of Bern, Bern, Switzerland</i>
09.2022 – 01.2024	Tutorial and Lab Course Assistant for the lectures "Physics for Medical Undergraduates" and "Physics for Biology and Pharmacy Undergraduates", <i>University of Bern, Bern, Switzerland</i>
01.2021	Co-Lecturer for the Ph.D. introductory course " Atmosphere, Ocean & Land 2021 " - Ph.D. Level Course, <i>Max Planck Institute for Biogeochemistry, Jena, Germany</i>
09.2019	Tutorial Assistant at Max Planck Institute Earth System Modelling Summer School 2019 - Ph.D. Level Course, <i>Max-Planck-Institute for Meteorology, Hamburg, Germany</i>
09.2011 – 08.2014	Tutorial Assistant for Introductory Chemistry 1 & 2 – Undergraduate Course, <i>ETH Zürich, Switzerland</i>
02.2012 – 01.2014	Tutorial Assistant for Systems Analysis, <i>ETH Zürich, Switzerland</i>

Supervision

01.07.2025 –	Advisor to Tamara Emmerichs, Postdoctoral Researcher, <i>Max-Planck-Institute for Meteorology, Hamburg, Germany</i>
07.2024 –	Supervisor to Lara Oxley , Ph.D. Candidate at <i>University of Bern, Bern Switzerland</i>
20.03.2026 – 08.2026	Supervisor to Valentin Lanz, Bachelor Thesis, <i>University of Bern, Bern, Switzerland</i>
08.2023 – 08.2025	Advisor to Yu Zhu, Ph.D. Candidate at <i>Max-Planck-Institute for Biogeochemistry, Jena, Germany</i>

Funding, Proposals and Awards

2026	Collaborator EU Horizon CLIMATE proposal : Shortlisted in Stage 2 review (11 Mio. EUR total requested)
2023	Awarded Ambizione Fellowship, 2022 Call (covering 799'000 CHF / 889'076 USD), covering own salary and a that of a PhD candidate for 4 years, with the project <i>"Past and Future Perturbation of the Pan-Arctic Nitrogen Cycle"</i>
2023	Awarded Travel Grant by University of Bern (covering 2'800 CHF / 3116 USD) for projects <i>"Cascading Impacts of Climate Change in the South Atlantic"</i> at <i>Universidade Federal de Santa Catarina, Florianopolis, Brazil</i>
2022	Contributor to Swiss National Computing Centre computing time proposals OVERSHOOT (large proposal) and AERA (small proposal)
2020	Contributor to computing time request proposal CRESCENDO
2017	Outstanding Student Poster, <i>10th International Carbon Dioxide Conference</i> , Interlaken, Switzerland

Selected Conference and Workshop Presentations

Lacroix et al.: Warming, Acidification, Greening and Deoxygenation in the Global Coastal Ocean, Ocean Science Meeting, 2026, Glasgow, Schotland (**Oral**)

Lacroix et al.: The R2O-MIP: The riverine influence on the global ocean CO₂ flux. *11th International Carbon Dioxide Conference*, 2024, Manaus, Brazil (**Oral**)

Lacroix et al.: Persistent High-Latitude Ocean Warming and Global Sea Level Rise Following Temporary Overshoots. *EGU General Assembly 2024*, (**Oral**)

Lacroix et al.: Enhanced Transports from Land-to-Ocean: Towards Boundless Assessments of Carbon and Nitrogen Cycles. *Oeschger Centre for Climate Change Seminar*, University of Bern, 6 Dec 2023. (**Invited, Oral**)

Lacroix et al.: Riverine Influence on the global ocean CO₂ flux. *R2O-MIP Workshop*, [SOCAT pCO₂ Workshop Series](#), Nov 2023, Oostende, Belgium. (**Invited Oral**)

Lacroix et al. Towards a boundless Earth system: Do riverine and coastal disruptions affect marine greenhouse gases and ecosystems at the global scale? Oct 2023, *Seminar Oceanografia*, University of Sao Paulo, Brazil. (**Invited Oral**)

Lacroix et al.: Temporal Disconnect of Seasonal Plant Nutrient Demand and Thaw Depth implies an Increasing Source of N₂O in High-Latitude Permafrost Ecosystems. *EGU General Assembly 2022*, Vienna, Austria (**Oral**)

Regnier, P. & **Lacroix, F.**: Coastal Ocean CO₂ fluxes and C cycle. *Ocean Carbon Budget Workshop* Jun. 2021, Online Workshop (**Invited, Oral**)

Lacroix, F. & Regnier, P: F_{nat} , more than just river-induced carbon outgassing? *Global Carbon Budget Ocean Workshop*, “Carbon Gaps”, Feb. 2021, Online Workshop (**Invited, Oral**)

Lacroix F. Ilyina, T., Mathis, M., Laruelle, G., Regnier, P: Short coastal water residence times support an efficient cross-shelf transfer of organic matter: Were pre-industrial continental shelves already a global CO₂ sink? *Ocean Sciences Meeting 2020* (**Oral**)

Selected Extra-Curricular Workshops

- 2019** C-CASCADES ITN Training Workshop – Entrepreneurial skills: Deltares, Delft, Netherlands.
- 2014** Avenues in the Tropics, *Barranquilla, Colombia*. Sustainable urban planning lab and workshop co-organized by Universidad del Norte (Colombia), ETH Zürich (Switzerland) and the World Bank (USA).

Memberships, Roles and Review

- Since 08.2026** Advisor to Schweizer Welt Atlas (Swiss geographic maps for primary and secondary schools)
- Since 08.2026** Reviewer for International Panel on Climate Change AR7, Chapter 9
- Since 08.2025** Steering Committee Member QUINCY Model (ICON-LAND subcomponent)
- Since 2019** EGU Membership
- 2016, 2025, 2026** AGU Membership
- 2024** Data Provider and Modeling Advisor for PROVIDE Climate Dashboard (<https://climate-risk-dashboard.iiasa.ac.at>)
- 2025** Co-Organizer PanArcN-Q-Arctic Field Campaign

Completed peer reviews for *Nature Climate Change*, *Nature Geoscience*, *Nature Communications*, *Communications Earth and Environment*, *Science Advances*, *Geophysical Research Letters*, *Biogeosciences*, *Earth System Dynamics*, *Cryosphere*, *Geophysical Model Development*, *Geophysical Research Letters*, *Journal of Advances in Modeling Earth Systems*, *Global Biogeochemical Cycles*

Publication List

In Review:

22. Samuel Lüthi, Fabrice Lacroix, Mireia Ginesta et al. Heat-mortality impacts under 1.5°C overshoot pathways, *In review for Nature Communications*. <https://doi.org/10.21203/rs.3.rs-9577848/v1>
21. Zhu, Y., **Lacroix, F.**, Zaehle, S.: Retrieving phenological parameters from remote sensing for modeling of permafrost-affected vegetation. *In review for JAMES*.
20. **Lacroix, F.**, Mathis, M., Regnier, R., Gomez, F., Rodrigues, R., Frölicher T.L., Le Grix, N., Takano, Y., Ilyina, T.: Rapid Warming, Acidification, Greening and Deoxygenation in the Global Coastal Ocean. *In Review, Nature Climate Change*, <https://www.researchsquare.com/article/rs-8095752/v1>.

Peer Reviewed:

2026

19. Lhardy et al. (incl. **Lacroix, F.**) : Weathering flux calculations from climate model outputs show a decrease in alkalinity fluxes to the ocean during the last deglaciation. *In print for Geophysical Research Letters*.
18. Emmerichs, T., **Lacroix, F.**, Brovkin, V., Zaehle, S., Gong, C., Zhu, Y., Sjogersten, S., Voigt, C., Steenberg Larsen, K., and Tuitila, E.-S.: Impacts of Shrub Coverage for Arctic Ecosystem Carbon Uptake and Storage, *In Print for Biogeoscience*, <https://doi.org/10.5194/egusphere-2026-1514>, 2026.
17. Seitz, J., Yajima, M, Zhu, Y., Joseph, L.S.K, Yang, J, Lacroix, F., Luo, Y., Schaumberger, A., Bahn, M., Zaehle, S., and Caldararu, S. (2026): A general physiologically driven representation of leaf turnover in grasslands in the QUINCY land surface model (revision: 974a6b7f). *Geophysical Model Development*. <https://doi.org/10.5194/egusphere-2025-5731>

2025

16. Schuster, L., Maussion, F., Rounce, D.R. et al. Incl. **Lacroix, F.** (2025): Irreversible glacier change and trough water for centuries after overshooting 1.5 °C. *Nat. Clim. Chang.* 15, 634–641 (2025). <https://doi.org/10.1038/s41558-025-02318-w>
15. Morée, A. L., **Lacroix, F.**, Cheung, W. W. L., and Frölicher, T. L. (2025): Long-term impacts of global temperature stabilization and overshoot on exploited marine species, *Biogeosciences*, 22, 1115–1133, <https://doi.org/10.5194/bg-22-1115-2025>

2024

14. Silvy, Y., Frölicher, T. L., Terhaar, et al., Incl. **Lacroix, F.**, (2024): AERA-MIP: emission pathways, remaining budgets, and carbon cycle dynamics compatible with 1.5 and 2 °C global warming stabilization, *Earth Syst. Dynam.*, 15, 1591–1628, <https://doi.org/10.5194/esd-15-1591-2024>
13. **Lacroix, F.**, Burger, F. A., Silvy, Y., Schleussner, C.-F., & Frölicher, T. L. (2024). Persistently elevated high-latitude ocean temperatures and global sea level following temporary overshoots. *Earth's Future*, 12, e2024EF004862. <https://doi.org/10.1029/2024EF004862>

12. Schleussner, C.F. et al., Incl. **Lacroix, F.** (2024). Overconfidence in climate overshoot. *Nature* **634**, 366–373. <https://doi.org/10.1038/s41586-024-08020-9>
11. Liu, M., Raymond, P.A., Lauerwald, R. et al., Incl. **Lacroix, F.** (2024). Global riverine land-to-ocean carbon export constrained by observations and multi-model assessment. *Nat. Geosci.* **17**, 896–904 (2024). <https://doi.org/10.1038/s41561-024-01524-z>
10. Ernst, Y., Archibald, S., Balzter, H., et al., Incl. **Lacroix, F.** (2024). The African regional greenhouse gases budget (2010–2019). *Global Biogeochemical Cycles*, 38, e2023GB008016. <https://doi.org/10.1029/2023GB008016>
9. Mathis, M., **Lacroix, F.**, Hagemann S., Nielsen, D., Ilyina, T., Schrum, C. (2024): Enhanced CO₂ uptake of the coastal ocean is dominated by biological carbon fixation. *Nature Climate Change*. <https://doi.org/10.21203/rs.3.rs-2928105/v1>
8. Resplandy, L., Hogikyan, A., Müller, J. D., et al., Incl. **Lacroix, F.** (2024). A synthesis of global coastal ocean greenhouse gas fluxes. *Global Biogeochemical Cycles*, 38, e2023GB007803. <https://doi.org/10.1029/2023GB007803>

2023

7. Schaller, J. et al. (incl. **Lacroix, F.**, 2023): Arctic soil CO₂ release during freeze-thaw cycles modulated by silicon and calcium. *Science of the Total Environment*, 870, 161943. <https://doi.org/10.1016/j.scitotenv.2023.161943>
6. Stimmler, P. et al. (incl. **Lacroix, F.**, 2023): Pan-Arctic soil element availability estimations. *Earth System Science Data*, 15, 1059–1075. <https://doi.org/10.5194/essd-15-1059-2023>

2022

5. Mathis, M., Logemann, K., Maerz, J., **Lacroix, F.**, Hagemann, S., Chegini, F., et al. (2022). Seamless integration of the coastal ocean in global marine carbon cycle modeling. *Journal of Advances in Modeling Earth Systems*, 14, e2021MS002789. <https://doi.org/10.1029/2021MS002789>
 4. **Lacroix, F.**, Zaehle, S., Caldararu, S., Schaller, J., Stimmler, P., Holl, D., Kutzbach, L., & Göckede, M. (2022). Mismatch of N release from the permafrost and vegetative uptake opens pathways of increasing nitrous oxide emissions in the high Arctic. *Global Change Biology*, 28, 5973– 5990. <https://doi.org/10.1111/gcb.16345>
- Dai, M., et al., Incl. **Lacroix, F.** (2022): Carbon Fluxes in the Coastal Ocean: Synthesis, Boundary Processes and Future Trend, *Annual Review of Earth and Planetary Sciences* 50(1). <https://doi.org/10.1146/annurev-earth-032320-090746>

2021

3. **Lacroix, F.**, Ilyina, T., Mathis, M., Laruelle, G. G., & Regnier, P. (2021): Historical increases in land-derived nutrient inputs may alleviate effects of a changing physical climate on the oceanic carbon cycle. *Global Change Biology*, 00, 1–23. <https://doi.org/10.1111/gcb.15822>
2. **Lacroix, F.**, Ilyina, T., Laruelle, G. G., & Regnier, P. (2021): Reconstructing the preindustrial coastal carbon cycle through a global ocean circulation model: Was the global continental shelf already both autotrophic

and a CO₂ sink?. *Global Biogeochemical Cycles*, 35, e2020GB006603. <https://doi.org/10.1029/2020GB006603>

2020

1. **Lacroix F.**, Ilyina T. & Hartmann, J. (2020): Oceanic CO₂ outgassing and biological production hotspots induced by pre-industrial river loads of nutrients and carbon in a global modeling approach, *Biogeosciences*, 17, 55–88. <https://doi.org/10.5194/bg-17-55-2020>

Non-peer reviewed

Lacroix, F., Liu M., Ma, M., et al. (2024). Model Protocol for River2Ocean (R2O) Model Intercomparison Project (MIP). Zenodo. <https://doi.org/10.5281/zenodo.15631028>

Lacroix F. (2023). Arctic Rivers tell Tales of Change. *Nature Geoscience*, 16, 760-761. <https://doi.org/10.1038/s41561-023-01248-6>

Selected Datasets

Lacroix, F. (2025). Observation Product and Model Changes in Coastal Ocean Stressors 1900-2022. Zenodo. <https://doi.org/10.5281/zenodo.17516000>

Lacroix, F., Ilyina, T., & Hartmann, J. (2025). Impacts of Biogeochemical River Inputs on Oceanic FCO₂ in MPIOM-HAMOCC [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.15631028>

Lacroix, F., Liu, M., Ma, M., Resplandy, L., Beusen, A., Hauck, J., Lennartz, S., Li, Y., Tian, H., & Regnier, P. (2024). Biogeochemical river inputs for global ocean models (RivR2O) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.13799103>

Lacroix, F., Burger, F., Silvy, Y., Schleussner, C.-F., & Frölicher, T. L. (2024). GFDL-ESM2M overshoot data [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.11091132>

Dashboards

Climate Risk Dashboard (<https://climate-risk-dashboard.iiasa.ac.at>)